

ACTIVITY 6: Testing the Account Class with Exceptions

Objective

Write a test class that handles exceptions thrown by the enhanced `Account` class.

Requirements

Create a test class `TestAccountExceptions.java` that:

1. Tests all scenarios with proper try-catch blocks
2. Displays informative messages when exceptions occur
3. Shows both success and failure cases
4. Demonstrates proper exception handling

Expected Output

When you run your test class, the output should look like this:

text

```
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ACCOUNT TEST WITH EXCEPTIONS
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```

>>> Test 1: Valid Account Creation

SUCCESS: Account #1001 | John Doe (25 yrs) | Savings | ₹1000.0 | Active | PIN: No

>>> Test 2: Invalid Age (under 18)

EXCEPTION: Customer must be at least 18 years old. Provided: 16

>>> Test 3: Invalid Account Type

EXCEPTION: Account type must be 'Savings' or 'Current'. Provided: Invalid

>>> Test 4: Minimum Balance on Creation

Creating Savings account with ₹300

EXCEPTION: Savings account requires minimum balance of ₹500.0. Provided: ₹300.0

>>> Test 5: Valid Deposit and Withdrawal

Account: Account #1005 | Alice Brown (30 yrs) | Current | ₹1000.0 | Active | PIN: No

Setting PIN 1234: SUCCESS

Depositing ₹500.0: SUCCESS

Balance after deposit: ₹1500.0

Withdrawing ₹200.0: SUCCESS

Balance after withdrawal: ₹1300.0

Account #1005 | Alice Brown (30 yrs) | Current | ₹1300.0 | Active | PIN: Yes

>>> Test 6: Invalid Deposit (Negative Amount)

Attempting to deposit ₹-100.0

EXCEPTION: Deposit amount must be positive. Provided: ₹-100.0

>>> Test 7: Insufficient Balance

Account: Account #1006 | Charlie Green (35 yrs) | Savings | ₹500.0 | Active | PIN: Yes

Attempting to withdraw ₹1000.0

EXCEPTION: Insufficient balance. Available: ₹500.0, Requested: ₹1000.0

>>> Test 8: Minimum Balance Violation

Account: Account #1007 | Diana Prince (28 yrs) | Savings | ₹1000.0 | Active | PIN: Yes

Attempting to withdraw ₹600.0

EXCEPTION: Cannot withdraw. Minimum balance of ₹500.0 required. Available after withdrawal: ₹400.0

>>> Test 9: Inactive Account Operations

Account: Account #1008 | Eve Wilson (32 yrs) | Current | ₹2000.0 | Active | PIN: No

Closing account: SUCCESS

Attempting to deposit ₹100.0 on closed account

EXCEPTION: Account is inactive. Please reopen the account or contact support.

Reopening account: SUCCESS

Depositing ₹100.0 after reopen: SUCCESS

Balance after deposit: ₹2100.0

>>> Test 10: PIN Verification

Account: Account #1009 | Frank Miller (40 yrs) | Savings | ₹1500.0 | Active | PIN: No

Setting PIN 1234: SUCCESS

Withdrawing ₹200.0 with correct PIN: SUCCESS

Balance: ₹1300.0

Attempting to withdraw ₹100.0 with incorrect PIN (9999)

EXCEPTION: Incorrect PIN

Attempting to withdraw ₹100.0 without PIN set

EXCEPTION: PIN not set for this account

>>> Test 11: All Accounts Summary

Account #1001 | John Doe (25 yrs) | Savings | ₹1000.0 | Active | PIN: No

Account #1005 | Alice Brown (30 yrs) | Current | ₹1300.0 | Active | PIN: Yes

Account #1006 | Charlie Green (35 yrs) | Savings | ₹500.0 | Active | PIN: Yes

Account #1007 | Diana Prince (28 yrs) | Savings | ₹1000.0 | Active | PIN: Yes

Account #1008 | Eve Wilson (32 yrs) | Current | ₹2100.0 | Active | PIN: No

Account #1009 | Frank Miller (40 yrs) | Savings | ₹1300.0 | Active | PIN: Yes

```
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TEST COMPLETED!
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```

Task

Write the complete `TestAccountExceptions` class in a file called `TestAccountExceptions.java`.

Handle all exceptions with try-catch blocks.